

Cyber Security



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INTRODUCTION

Cyber refers to anything related to computers, information technology, networks, and the internet. It represents the digital environment where people communicate, share information, and perform various online activities through interconnected systems and networks. Cyber Security is the practice of protecting computers, networks, software, and data from unauthorized access, attacks, damage, or theft. It helps ensure the confidentiality, integrity, and availability of information while safeguarding individuals and organizations from cyber threats. (Oxford Reference, 2023).



Cyber Crime

- Cyber crime refers to any illegal activity that is carried out using computers, networks, or the internet, where the computer may be the target, tool, or both. It includes activities such as hacking, identity theft, online fraud, and data breaches.
- Cyber crime is defined as criminal activities involving computers and digital networks (UNODC, 2021).



Major Cyber Security Problem



Advantages of Cyber Security



Data Protection

Protects personal and business information from hackers.



Business Security

Protects company data and builds trust.



Safe Online Transactions

Ensures secure online banking and payments.



Protection from Cyber Attacks

Prevents viruses, malware, and hacking.



User Confidence

Increases trust in digital platforms.



Real-Life Example of Cyber Attack



LinkedIn (2012)

Over 160 million user passwords were stolen and leaked on the dark web.



Equifax (2017)

Personal data of 147 million people was stolen due to a software vulnerability.



Uber (2022)

An 18-year-old hacker used social engineering to access Uber's internal cloud systems.



Adobe (2013)

Stole data for 38 million users, including encrypted credit card information,



Yahoo (2013-14)

The largest breach in history, affecting 3 billion user accounts.



Bangladesh Bank (2016)

Hackers attempted to steal \$81 million using the SWIFT network.



SolarWinds (2020)

A sophisticated supply chain attack that compromised major U.S. government agencies.

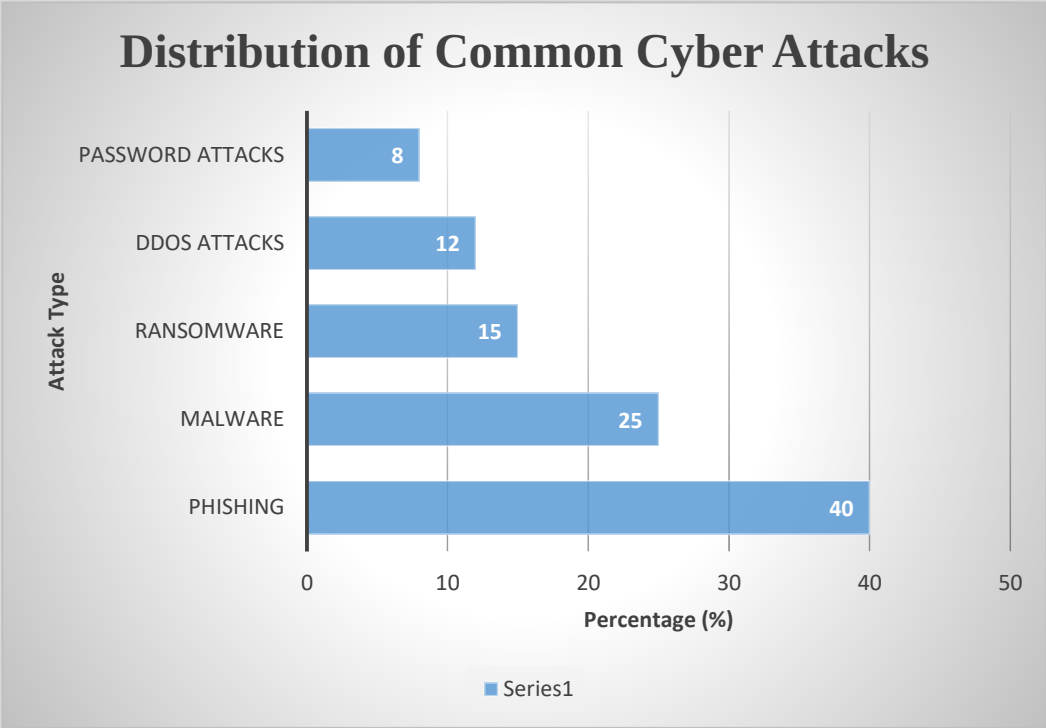


Target (2013 Breach)

Target experienced a major breach where around 40 million credit and debit card details of customers were stolen.

Statistical Analysis of Cyber Threats

Cyber Attack Type	Percentage (%)
Phishing	40
Malware	25
Ransomware	15
DDoS Attacks	12
Password Attacks	8



According to the dataset, Phishing attacks account for the highest percentage (40%), followed by Malware (25%). This chart highlights the most common cyber security threats and emphasizes the importance of effective security measures.

Conclusion

Cyber Security is essential for protecting computers, networks, and data from cyber threats. It helps ensure privacy, secure communication, and safe online activities. As technology continues to grow, strong cyber security practices are necessary to create a safer digital environment.

By increasing awareness and following security measures, individuals and organizations can reduce cyber risks and use technology more safely.



AI Usage Log

Tool Used	Purpose	Example Prompt (Short)	Output Used	Modification Done	Verification Method	Who Used It
ChatGPT & Gemini	Presentation Structure, Design & Data Integration	"Generate a pro-tier cybersecurity slide deck structure with technical definitions."	Presentation layout & high-end formatting suggestions	Lead Architect: Custom-engineered all professional slide layouts, visual themes, and integrated the Excel statistical chart	Manually verified chart percentages and logical presentation flow.	2026000000195
Grok AI	Data Acquisition	"Fetch 2024 statistics on phishing and ransomware breaches."	Threat percentages & attack numbers	Summarized historical breach data for the case study sections.	Cross-checked with IBM Security & Kaspersky reports.	2026000000189
NotebookLM	Literature Review	"Summarize mitigation steps from NIST PDF."	NIST framework summaries	Extracted core points for strategic defense.	Verified with official NIST documentation.	2026000000185
NotebookLM	Academic Research	"Format NIST bibliography."	Structured reference lists	Formatted final academic bibliography.	Cross-checked with textbook bibliography.	2026000000205

References

- National Institute of Standards and Technology (NIST). (2020). Framework for Improving Critical Infrastructure Cybersecurity. U.S. Department of Commerce. <https://www.nist.gov>
- United Nations Office on Drugs and Crime (UNODC). (2021). Comprehensive Study on Cybercrime. United Nations. <https://www.unodc.org>
- Oxford Reference. (2023). A Dictionary of Elements of Cyber Security. Oxford University Press.
- IBM Security. (2024). Cost of a Data Breach Report 2024. IBM Corporation. <https://www.ibm.com/reports/data-breach>
- Reuters. (2016). How Swifts and Hackers Aligned to Rob Bangladesh Bank of \$81 Million. Reuters News. <https://www.reuters.com>

Thank



You